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United States Patent	12417988
Kind Code	B2
Date of Patent	September 16, 2025
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Interposer, method for fabricating the same, and semiconductor package having the same

Abstract

An interposer according to an embodiment of the present invention includes a base layer having opposite first and second surfaces, a wiring structure on the first surface of the base layer, an interposer protective layer disposed on the second surface of the base layer and having a pad recess with a lower surface of the interposer protective layer positioned at a first vertical level and a bottom surface of the pad recess positioned at a second vertical level that is higher than the first vertical level, an interposer pad of which a portion fills the pad recess of the interposer protective layer and the remaining portion protrudes from the interposer protective layer, and an interposer through electrode extending through the base layer and the interposer protective layer to the interposer pad, the interposer through electrode electrically connecting the wiring structure to the interposer pad.

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Appl. No.: 18/643474

Filed: April 23, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240274553 A1	Aug. 15, 2024

Foreign Application Priority Data

KR	10-2021-0034855	Mar. 17, 2021
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Related U.S. Application Data

continuation parent-doc US 17528954 20211117 US 11984415 child-doc US 18643474

Publication Classification

Int. Cl.: **H01L23/498** (20060101); **H01L21/48** (20060101); **H01L23/00** (20060101); **H01L23/544** (20060101); **H01L25/10** (20060101); **H01L25/18** (20230101); H01L23/48 (20060101)

U.S. Cl.:

CPC **H01L24/02** (20130101); **H01L21/486** (20130101); **H01L23/49838** (20130101); **H01L23/544** (20130101); **H01L24/03** (20130101); **H01L24/05** (20130101); **H01L24/06** (20130101); **H01L25/105** (20130101); **H01L25/18** (20130101); H01L23/481 (20130101); H01L23/49816 (20130101); H01L23/49827 (20130101); H01L24/04 (20130101); H01L2223/54426 (20130101); H01L2224/0213 (20130101); H01L2224/02141 (20130101); H01L2224/02145 (20130101); H01L2224/0215 (20130101); H01L2224/03019 (20130101); H01L2224/03462 (20130101); H01L2224/0401 (20130101); H01L2224/05019 (20130101); H01L2224/05025 (20130101); H01L2224/05082 (20130101); H01L2224/0603 (20130101); H01L2224/06182 (20130101)

Field of Classification Search

CPC: H01L (24/94); H01L (23/841); H01L (23/49838)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a continuation of U.S. patent application Ser. No. 17/528,954 filed on Oct. 11, 2021, which is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0034855, filed on Mar. 17, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(2) The present disclosure relates to an interposer electrically connecting a semiconductor chip to a main board, a method of manufacturing the interposer, and a semiconductor package having the interposer.

(3) As miniaturization, multifunctionality, and high performance of electronic products are required, light and thin, high integration, high performance, and high speed of semiconductor packages are also required. Accordingly, the demand for a semiconductor package capable of implementing a system having a high memory bandwidth is increasing. Because the memory bandwidth is proportional to the data transmission speed and the number of data transmission lines, the memory bandwidth may be increased by increasing the memory operation speed or increasing the number of data transmission lines. Accordingly, in order to increase the number and density of connection bumps attached to connection pads of a semiconductor chip, a semiconductor package using an interposer has been introduced.

SUMMARY

(4) Embodiments of the present invention provides an interposer capable of increasing a memory bandwidth and a semiconductor package having the interposer.

(5) In order to solve the above technical problem, embodiments of the present invention provides the following interposer, and a semiconductor package having the same.

(6) An interposer according to an embodiment of the present invention includes: a base layer having opposite first and second surfaces; a wiring structure on the first surface of the base layer; an interposer protective layer disposed on the second surface of the base layer and having a pad recess with a lower surface of the interposer protective layer positioned at a first vertical level and a bottom surface of the pad recess positioned at a second vertical level that is higher than the first vertical level; an interposer pad of which a portion fills the pad recess of the interposer protective layer and the remaining portion protrudes from the interposer protective layer; and an interposer through electrode extending through the base layer and the interposer protective layer to the interposer pad, the interposer through electrode electrically connecting the wiring structure to the interposer pad.

(7) An interposer according to an embodiment of the present invention includes: a base layer having opposite first and second surfaces; a wiring structure on the first surface of the base layer; a lower interposer protective layer formed on the second surface of the base layer; an upper interposer protective layer formed on the lower interposer protective layer; an interposer pad of which a first portion filling a pad recess extending from an upper surface of the upper interposer

protective layer into the lower interposer protective layer and a second portion protruding from the upper interposer protective layer are integrally formed; an interposer through electrode extending through the base layer and the lower interposer protective layer to the interposer pad up to the same vertical level as an upper surface of the upper interposer protective layer, the interposer through electrode electrically connecting the wiring structure to the interposer pad; and an alignment key penetrating the upper interposer protective layer and extending into the lower interposer protective layer.

(8) A semiconductor package according to an embodiment of the present invention includes: an interposer including a base layer having a first surface and a second surface opposite to each other, a wiring conductive layer disposed on the first surface of the base layer, an inter-wiring insulating layer disposed on the first surface of the base layer and surrounding the wiring conductive layer, a plurality of micro pads electrically connected to the wiring conductive layer, an interposer protective layer covering the second surface of the base layer, a plurality of interposer pads each including a first portion disposed below a top surface of the interposer protective layer and a second portion protruding from the interposer protective layer, and a plurality of interposer through electrodes electrically connecting the wiring conductive layer to the plurality of interposer pads; at least one sub-semiconductor package mounted on the first surface of the base layer to be electrically connected to some of the plurality of micro-pads, and including a first semiconductor chip and a plurality of second semiconductor chips sequentially stacked; and a third semiconductor chip spaced apart from the at least one sub-semiconductor package in a horizontal direction and mounted on the first surface of the base layer to be electrically connected to some other of the plurality of micro-pads.

(9) An interposer manufacturing method according to an embodiment of the present invention includes: preparing a base layer having a first surface and a second surface opposite to each other, and an interposer through electrode penetrating the base layer and protruding from the second surface of the base layer; forming an interposer protective layer covering the second surface of the base layer and having an upper surface at the same vertical level as an upper surface of the interposer through electrode; removing a portion of the interposer protective layer to form a pad recess in which an upper portion of the interposer through electrode is exposed and a portion of the interposer protective layer is exposed on a bottom surface; forming an interposer pad filling the pad recess of the interposer protective layer and protruding from the interposer protective layer; and forming a rear wiring protective layer having a terminal opening exposing a portion of the interposer pad and covering the remaining portion of the interposer pad and the interposer protective layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a cross-sectional view showing an interposer according to an embodiment of the present invention;

(3) FIG. 2 is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention;

(4) FIG. 3 is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention;

(5) FIG. 4 is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention;

(6) FIG. 5 is an enlarged cross-sectional view illustrating portions of an interposer according to an

embodiment of the present invention;

(7) FIGS. 6A to 6G are cross-sectional views illustrating a method of manufacturing an interposer according to an embodiment of the present invention in stages;

(8) FIGS. 7A to 7D are cross-sectional views illustrating a method of manufacturing an interposer according to an embodiment of the present invention in stages; and

(9) FIG. 8 is a cross-sectional view illustrating a semiconductor package having an interposer according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(10) FIG. 1 is a cross-sectional view showing an interposer according to an embodiment of the present invention.

(11) Referring to FIG. 1, an interposer 500 includes a base layer 510, a plurality of interposer through electrodes 520 penetrating the base layer 510 from a first surface 512 to a second surface 514 of the base layer 510, a wiring structure 530 disposed on the first surface 512 of the base layer 510, a plurality of micro pads 540 on the first surface 512 of the base layer 510 and electrically connected to the wiring structure 530, an interposer protective layer 550 covering the second surface 514 of the base layer 510, a plurality of interposer pads 570 partially buried in the interposer protective layer 550, and a rear wiring protective layer 560 covering the plurality of interposer pads 570. The interposer protective layer 550 may expose a portion of the plurality of interposer pads 570. The interposer 500 may further include a plurality of interposer connection terminals 580 respectively attached to the plurality of interposer pads 570.

(12) As used herein, components described as being “electrically connected” are configured such that an electrical signal can be transferred from one component to the other (although such electrical signal may be attenuated in strength as it transferred and may be selectively transferred).

(13) The base layer 510 may include or be formed of a semiconductor material, glass, ceramic, or plastic. For example, the base layer 510 may include or be formed of silicon. In some embodiments, the base layer 510 may include or may be a silicon semiconductor substrate.

(14) Each of the plurality of interposer through electrodes 520 may include or be formed of a conductive plug passing through the base layer 510 and a conductive barrier film surrounding the conductive plug. The conductive plug may include or be formed of Cu or W, and the conductive barrier film may include or be formed of a metal or a conductive metal nitride. The conductive plug may have a cylindrical shape, and the conductive barrier film may have a cylindrical shape surrounding a sidewall of the conductive plug. A plurality of via insulating films may be interposed between the base layer 510 and the plurality of interposer through electrodes 520 to surround sidewalls of the plurality of interposer through electrodes 520. The plurality of via insulating films may prevent contact between the base layer 510 and the plurality of interposer through electrodes 520. The via insulating film may include or be formed of an oxide film, a nitride film, a carbon film, a polymer, or a combination thereof.

(15) It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point of contact.

(16) In FIG. 1, the first surface 512 and the second surface 514 are respectively shown as being on the lower side and the upper side of the base layer 510, and in the semiconductor package 1000 having the interposer 500 in FIG. 8, the first surface 512 and the second surface 514 may be referred to as an upper surface and a lower surface, respectively. The interposer 500 shown in FIG. 1 and the interposer 500 included in the semiconductor package 1000 shown in FIG. 8 are upside down with respect to each other.

(17) The interposer 500 may further include an alignment key AK disposed on the second surface 514 of the base layer 510. The alignment key AK may be configured by an interface between the

interposer protective layer **550** and the rear wiring protective layer **560** extending into the interposer protective layer **550**.

(18) FIG. **2** is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention. For example, FIG. **2** is an enlarged cross-sectional view illustrating regions IIA and IIB of FIG. **1**.

(19) Referring to FIGS. **1** and **2** together, the interposer **500** may include a base layer **510**, a plurality of interposer through electrodes **520**, a wiring structure **530**, a plurality of micro pads **540**, an interposer protective layer **550**, a plurality of interposer pads **570**, and a rear wiring protective layer **560**.

(20) The wiring structure **530** may include a wiring conductive layer **532** and an inter-wiring insulating layer **534** surrounding the wiring conductive layer **532**. The wiring conductive layer **532** may electrically connect the plurality of interposer through electrodes **520** to the plurality of micro pads **540**. The wiring conductive layer **532** may include or be formed of a metal material such as aluminum, copper, or tungsten. The inter-wiring insulating layer **534** may include or be formed of silicon oxide. In some embodiments, the inter-wiring insulating layer **534** may be formed of tetraethyl orthosilicate (TEOS). In some other embodiments, the inter-wiring insulating layer **534** may be made of an insulating material having a lower dielectric constant than silicon oxide, for example, an ultra low k (ULK) film having an ultra low dielectric constant K of about 2.2 to about 2.4.

(21) The wiring conductive layer **532** may include a plurality of wiring patterns constituting a plurality of wiring layers and a plurality of wiring vias connecting wiring patterns disposed on different wiring layers. The wiring layer may be an electrical path extending in a plane, and different wiring layers may be electrical paths horizontally extending in planes at different vertical levels, respectively.

(22) A plurality of pad wiring patterns **536** electrically connected to the wiring conductive layer **532** and disposed on the wiring structure **530** may be between the wiring conductive layer **532** and the plurality of micro pads **540**. For example, the plurality of pad wiring patterns **536** may include or be formed of copper, nickel, or a copper alloy. A front wiring protective layer **538** may expose a portion of the plurality of pad wiring patterns **536** and cover the wiring structure **530** and a portion of the plurality of pad wiring patterns **536**. The front wiring protective layer **538** may include or be formed of epoxy or polyimide.

(23) Each of the plurality of micro pads **540** may include a front under bump metallurgy (UBM) layer **542** in contact with the pad wiring pattern **536** and a chip connection terminal **544** in contact with the front UBM layer **542**. The plurality of micro pads **540** may be formed of a conductive material, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Tin), gold (Au), or solder, but are limited thereto.

(24) The interposer protective layer **550** may include a lower interposer protective layer **552** covering the second surface **514** of the base layer **510** and an upper interposer protective layer **554** covering the lower interposer protective layer **552**. The interposer protective layer **550** may be formed of an inorganic material. For example, the interposer protective layer **550** may be formed of silicon oxide, silicon nitride, or a stacked structure of silicon oxide and silicon nitride. For example, the lower interposer protective layer **552** may be made of oxide, and the upper interposer protective layer **554** may be made of nitride. In some embodiments, the lower interposer protective layer **552** may be made of silicon oxide, and the upper interposer protective layer **554** may be made of silicon nitride.

(25) The interposer protective layer **550** may have a pad recess **570R** formed on an upper surface of the upper interposer protective layer **550**. For example, side walls of the pad recess **570R** may extend from an upper surface of the upper interposer protective layer **554** toward the base layer **510**. A portion of the lower interposer protective layer **552** may be exposed on the bottom surface of the pad recess **570R**. The pad recess **570R** may extend from the upper surface of the upper

interposer protective layer **554** toward the base layer **510**, but may not completely penetrate the interposer protective layer **550**. In some embodiments, the pad recess **570R** may penetrate the upper interposer protective layer **554**, but may penetrate an upper portion of the lower interposer protective layer **552** and may not penetrate a lower portion thereof.

(26) The lower interposer protective layer **552** and the upper interposer protective layer **554** may have a first thickness **T1** and a second thickness **T2** as a maximum thickness, respectively. For example, the first thickness **T1** of a portion of the lower interposer protective layer **552** and the second thickness **T2** of a portion of the upper interposer protective layer **554** that do not vertically overlap with the plurality of pad recesses **570R** and the plurality of interposer pads **570** may be substantially constant. In some embodiments, the first thickness **T1** may be greater than the second thickness **T2**. For example, the second thickness **T2** may be about 0.5 μm to about 1.5 μm , and the first thickness **T1** may be greater than the second thickness **T2**, but may be about 2.5 μm or less.

(27) Terms such as “constant,” “same,” “equal,” “planar,” or “coplanar,” as used herein when referring to orientation, layout, location, shapes, sizes, compositions, amounts, or other measures do not necessarily mean an exactly identical orientation, layout, location, shape, size, composition, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, compositions, amounts, or other measures within acceptable variations that may occur, for example, due to manufacturing processes. The term “substantially” may be used herein to emphasize this meaning, unless the context or other statements indicate otherwise. For example, items described as “substantially constant,” “substantially the same,” “substantially equal,” or “substantially planar,” may be exactly constant, the same, equal, or planar, or may be constant, the same, equal, or planar within acceptable variations that may occur, for example, due to manufacturing processes.

(28) A lower surface of the lower interposer protective layer **552** may be at a first vertical level **LV1**, and a bottom surface of the pad recess **570R** may be at a second vertical level **LV2** that is higher than the first vertical level **LV1**. The upper surface of the lower interposer protective layer **552** and the lower surface of the upper interposer protective layer **554** may contact each other at a third vertical level **LV3** that is higher than the second vertical level **LV2**. An upper surface of the upper interposer protective layer **554** may be at a fourth vertical level **LV4** that is higher than the third vertical level **LV3**.

(29) Each of the plurality of interposer pads **570** may include a seed layer **572** covering the bottom surface and side surfaces of the pad recess **570R**, e.g., the surface of the interposer protective layer **550** exposed in the pad recess **570R**, and a conductive pad layer **574** on the seed layer **572**. The seed layer **572** may conformally cover the bottom surface and side surfaces of the pad recess **570R**. At the side surface of the pad recess **570R**, an uppermost end of the seed layer **572** may be at the fourth vertical level **LV4**.

(30) Each of the plurality of interposer pads **570** may fill the pad recess **570R** and protrude more outward/upward than the upper surface of the interposer protective layer **550**, e.g., the upper surface of the upper interposer protective layer **554**. For example, upper surfaces of the plurality of interposer pads **570** may be at a vertical level that is higher than the fourth vertical level **LV4**. Lower surfaces of the plurality of interposer pads **570** may be at the second vertical level **LV2**.

(31) The plurality of interposer through electrodes **520** may extend into the plurality of interposer pads **570**. Upper surfaces of the plurality of interposer through electrodes **520** may be at the fourth vertical level **LV4**. For example, the upper surface of the plurality of interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554** may be at the fourth vertical level **LV4**. For example, the upper surface of the plurality of interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554** at the same vertical level.

(32) The lowermost surfaces of the plurality of interposer pads **570**, e.g., a portion excluding the portion in contact with the plurality of interposer through electrodes **520**, among the lower surfaces of the plurality of interposer pads **570** may all be in contact with the lower interposer protective

layer **552**. For example, the lower surfaces of the plurality of interposer pads **570** may be covered by and/or in contact with the plurality of interposer through electrodes **520** and the lower interposer protective layer **552**.

(33) Among the plurality of interposer through electrodes **520**, a portion above the first vertical level LV1 at which the upper surface of the base layer **510** is positioned may be surrounded by the lower interposer protective layer **552** and the plurality of interposer pads **570**. Among the plurality of interposer through electrodes **520**, the upper surfaces and side surfaces of the uppermost portions among the portions above the upper surface of the base layer **510** may be surrounded by a plurality of interposer pads **570**, and side surfaces of the remaining portions (e.g., the other portions) among the portions above the upper surface of the base layer **510** may be surrounded by the lower interposer protective layer **552**.

(34) Each of the plurality of interposer pads **570** may have a third thickness T3 as a maximum thickness. The third thickness T3 may be about 2 μm to about 7 μm . A portion below the fourth vertical level LV4 of the plurality of interposer pads **570**, e.g., a portion buried in the interposer protective layer **550** (e.g., a portion below the top surface of the interposer protective layer **550**), may have a fourth thickness T4, and another portion which is not buried in the interposer protective layer **550** and is above the fourth vertical level LV4, e.g., a portion protruding from/above the upper surface of the interposer protective layer **550** may have a fifth thickness T5. A portion buried in the interposer protective layer **550** (e.g., a portion below the top surface of the interposer protective layer **550**) among the plurality of interposer pads **570** may be referred to as a first portion, and a portion of the interposer pads **570** protruding from/above the upper surface of the interposer protective layer **550** may be referred to as a second portion. The first portion and the second portion may form an integral body of an interposer pad **570**. In some embodiments, the fifth thickness T5 may be greater than the fourth thickness T4. For example, the fifth thickness T5 may be greater than the fourth thickness T4, but may have a value of about 4 μm or less.

(35) The rear wiring protective layer **560** may have a terminal opening **5600** exposing a portion of the plurality of interposer pads **570**, and cover the remaining portions (e.g., the other portions) of the plurality of interposer pads **570** and the interposer protective layer **550**. The rear wiring protective layer **560** may cover all side surfaces of portions of the plurality of interposer pads **570** protruding from the upper surface of the interposer protective layer **550** and cover a portion of an upper surface of the interposer pads **570**, but may have a terminal opening **5600** not covering and exposing the remaining portion (e.g., the other portion) of the upper surface of the interposer pads **570**. For example, the rear wiring protective layer **560** may contact portions of side surfaces of the interposer pads **570** protruding from the upper surface of the interposer protective layer **550**. The rear wiring protective layer **560** may be made of an organic material. For example, the rear wiring protective layer **560** may be formed of a polymer material. In some embodiments, the rear wiring protective layer **560** may be formed of a photo imageable dielectric (PID) such as polyimide.

(36) The rear wiring protective layer **560** may have a sixth thickness T6 on the interposer protective layer **550**. For example, on the interposer protective layer **550**, the lowest thickness of the rear wiring protective layer **560** may be the sixth thickness T6, and the thickness of the rear wiring protective layer **560** adjacent to the plurality of interposer pads **570** on the interposer protective layer **550** may be greater than the sixth thickness T6. The sixth thickness T6 may be greater than the fifth thickness T5. For example, the sixth thickness T6 may be about 3 μm to about 4.5 μm . The highest thickness of the rear wiring protective layer **560** on the plurality of interposer pads **570** may be a seventh thickness T7, and the lowest thickness of the rear wiring protective layer **560** on the interposer pads **570** may be an eighth thickness T8. For example, the seventh thickness T7 may be about 3 μm to about 4.5 μm , and the eighth thickness T8 may be about 1 μm to about 2.5 μm .

(37) Each of the plurality of interposer connection terminals **580** may include a rear UBM layer **582** on the interposer pad **570**, and an interposer conductive cap **584** on the rear UBM layer **582**.

(38) The rear UBM layer **582** may cover/contact a portion of the upper surface of the interposer

pad **570** exposed by the terminal opening **5600** and a portion of the rear wiring protective layer **560** adjacent thereto. In some embodiments, the rear UBM layer **582** may conformally cover a portion of the upper surface of the interposer pad **570** exposed by the terminal opening **5600** and a portion of the rear wiring protective layer **560** adjacent thereto, with a ninth thickness **T9**. The ninth thickness **T9** may be less than the seventh thickness **T7**. For example, the ninth thickness **T9** may be about 2 μm to about 3 μm . The rear UBM layer **582** may have a concave upper surface corresponding to the terminal opening **5600**.

(39) The interposer conductive cap **584** may cover all or most of the upper surface of the rear UBM layer **582**. In some embodiments, the interposer connection terminal **580** may be formed of a conductive material, for example, copper (Cu), aluminum (Al), silver (Ag), tin (Tin), gold (Au), or solder, but the present invention is not limited thereto. The interposer connection terminal **580** may include a multilayer or a single layer. In some embodiments, the interposer conductive cap **584** may be made of SnAg.

(40) The interposer protective layer **550** may further have an alignment recess **550R** extending from the upper surface of the upper interposer protective layer **554** toward the base layer **510**. A portion of the lower interposer protective layer **552** may be exposed on the bottom surface of the alignment recess **550R**. The alignment recess **550R** may extend from the upper surface of the upper interposer protective layer **554** toward the base layer **510**, but may not completely penetrate the interposer protective layer **550**. In some embodiments, the alignment recess **550R** may penetrate the upper interposer protective layer **554**, but may penetrate only an upper portion of the lower interposer protective layer **552** and may not penetrate a lower portion thereof.

(41) The alignment recess **550R** may be filled by a rear wiring protective layer **560**. The alignment recess **550R** may be disposed to be spaced apart from the pad recess **570R**. The alignment recess **550R** may constitute/form an alignment key **AK**. The alignment key **AK** may be configured/formed by an interface between the interposer protective layer **550** and the rear wiring protective layer **560** extending into the interposer protective layer **550**. For example, the alignment key **AK** may be formed of the bottom surface and side surface of the alignment recess **550R**. For example, the alignment key **AK** may extend vertically and/or horizontally along the bottom surface and side surfaces of the alignment recess **550R**. The bottom surface of the alignment recess **550R** and the bottom surface of the pad recess **570R** may be at the second vertical level **LV2**. For example, the bottom surface of the alignment recess **550R** and the bottom surface of the pad recess **570R** may be at the same vertical level.

(42) The interposer **500** according to an embodiment of the present invention may include a plurality of interposer pads **570** that are partially buried in the interposer protective layer **550**, e.g., in the pad recess **570R**. Accordingly, the plurality of interposer pads **570** may be formed thickly while the height of a portion protruding from/above the interposer protective layer **550** is relatively low.

(43) Therefore, even when the wiring conductive layer **532** constitutes a relatively large number of wiring layers and thus, the thickness of the wiring conductive layer **532** and the volume occupied by the wiring conductive layer **532** in the interposer **500** increase, the thickness of the plurality of interposer pads **570** may be increased because the height of the protruding portion of the interposer pads **570** is low, so that by adjusting the ratio of the plurality of interposer pads **570** and the wiring conductive layer **532** disposed on the upper and lower portions of the interposer **500**, a warpage phenomenon may be prevented from occurring in the interposer **500**. For example, the interposer pad configuration/structure of the present disclosure may be beneficial to reduce or prevent warpage of interposers.

(44) FIG. 3 is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention. Specifically, each of IIIA and IIIB of FIG. 3 is an enlarged cross-sectional view showing a portion corresponding to the regions IIA and IIB of FIG. 1, and the same reference numerals as in FIG. 1 indicate the same components, and repeated descriptions

thereof may be omitted.

(45) Referring to FIGS. 1 and 3 together, an interposer **500a** may include a base layer **510**, a plurality of interposer through electrodes **520**, a wiring structure **530**, a plurality of micro pads **540**, an interposer protective layer **550a**, a plurality of interposer pads **570**, and a rear wiring protective layer **560**.

(46) The interposer protective layer **550a** may include a lower interposer protective layer **552a** covering the second surface **514** of the base layer **510** and an upper interposer protective layer **554a** covering the lower interposer protective layer **552a**. The interposer protective layer **550a** may be formed of an inorganic material. For example, the lower interposer protective layer **552a** may be made of oxide, and the upper interposer protective layer **554a** may be made of nitride. The lower interposer protective layer **552a** may protrude upward from the lower surface of the upper interposer protective layer **554a** while surrounding the side surfaces of the plurality of interposer through electrodes **520**, and may be in contact with lower surfaces of the plurality of interposer pads **570**. For example, the lower interposer protective layer **552a** may extend upward along lateral surfaces of the interposer through electrodes **520** so that the lower interposer protective layer **552a** may penetrate the upper interposer protective layer **554a**.

(47) The interposer protective layer **550a** may have a pad recess **570Ra** formed on an upper surface of the interposer protective layer **550a**. For example, side walls of the pad recess **570Ra** may extend from an upper surface of the upper interposer protective layer **554a** toward the base layer **510**. A portion of the upper interposer protective layer **554a** and a portion of the lower interposer protective layer **552a** may be exposed on the bottom surface of the pad recess **570Ra**. A portion of the lower interposer protective layer **552a** exposed on the bottom surface of the pad recess **570Ra** may surround the plurality of interposer through electrodes **520**. The pad recess **570Ra** may extend from the upper surface of the upper interposer protective layer **554a** toward the base layer **510**, but may not completely penetrate the interposer protective layer **550a**. In some embodiments, the pad recess **570Ra** may penetrate an upper portion of the upper interposer protective layer **554a** and may not penetrate a lower portion of the upper interposer protective layer **554a**.

(48) The lower interposer protective layer **552a** may have a first thickness **T1a** below the upper interposer protective layer **554a**, and the upper interposer protective layer **554a** may have a second thickness **T2a** as a maximum thickness. In some embodiments, the first thickness **T1a** may be less than the second thickness **T2a**. For example, the second thickness **T2a** may be about 0.5 μm to about 1.5 μm , and the first thickness **T1a** may be less than the second thickness **T2a**, but may be about 0.3 μm or more.

(49) A portion of the lower interposer protective layer **552a** that contacts the lower surfaces of the plurality of interposer pads **570** and surrounds side surfaces of the plurality of interposer through electrodes **520** may have a thickness greater than the first thickness **T1a** in the vertical direction, and the remaining portion (e.g., the other portion) of the lower interposer protective layer **552a** may have a first thickness **T1a** in a vertical direction. In some embodiments, a portion of the lower interposer protective layer **552a** that contacts the lower surfaces of the plurality of interposer pads **570** and surrounds side surfaces of the plurality of interposer through electrodes **520** may have the first thickness **T1a** in a horizontal direction on side surfaces of the plurality of interposer through electrodes **520**.

(50) For example, a portion of the lower interposer protective layer **552a** that overlaps with the upper interposer protective layer **554a** in the vertical direction may have a first thickness **T1a** that is substantially constant in the vertical direction, and portions of the upper interposer protective layer **554a** that do not vertically overlap with the plurality of pad recesses **570Ra** and the plurality of interposer pads **570** may have a second thickness **T2a** that is substantially constant. For example, the maximum thickness of the lower interposer protective layer **552a** may be the sum of the thickness of the upper interposer protective layer **554a** and the first thickness **T1a** at the portion overlapping the plurality of pad recesses **570Ra** and the plurality of interposer pads **570** in the

vertical direction. For example, the lower interposer protective layer **552a** may have a maximum thickness at a portion vertically overlapping the interposer pads **570**, and the maximum thickness of the lower interposer protective layer **552a** may be the same as a sum of a thickness of the upper interposer protective layer **554a** in a portion vertically overlapping both of an interposer pad **570** and the lower interposer protective layer **552a** and a thickness of the lower interposer protective layer **552a** in a portion vertically overlapping both of the interposer pad **570** and the upper interposer protective layer **554a**.

(51) A lower surface of the lower interposer protective layer **552a** may be at a first vertical level **LV1**, and a bottom surface of the pad recess **570Ra** may be at a second vertical level **LV2** that is higher than the first vertical level **LVI**. A portion of the upper surface of the lower interposer protective layer **552a** in contact with the lower surface of the upper interposer protective layer **554a** may be at a third vertical level **LV3a** that is lower than the second vertical level **LV2**. The upper surface of the upper interposer protective layer **554a** may be at a fourth vertical level **LV4** that is higher than the second vertical level **LV2**.

(52) The plurality of interposer through electrodes **520** may extend into the plurality of interposer pads **570**. For example, an upper portion of the interposer through electrodes **520** may horizontally overlap and be surrounded by the interposer pads **570**. The upper surface of the plurality of interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554a** may be at the fourth vertical level **LV4**, e.g., at the same vertical level.

(53) The lowermost surfaces of the plurality of interposer pads **570**, e.g., a portion of the lower surface of the plurality of interposer pads **570** excluding a portion in contact with the plurality of interposer through electrodes **520**, may be in contact with the upper interposer protective layer **554a** and the lower interposer protective layer **552a**. For example, among the lowermost surfaces of the plurality of interposer pads **570**, a portion adjacent to the plurality of interposer through electrodes **520** may be in contact with the lower interposer protective layer **552a**, and the remaining portions (e.g., the other portions) may be in contact with the upper interposer protective layer **554a**. For example, the lower surfaces of the plurality of interposer pads **570** may be all covered by the plurality of interposer through electrodes **520**, the upper interposer protective layer **554a**, and the lower interposer protective layer **552a**.

(54) Among the plurality of interposer through electrodes **520**, a portion above the first vertical level **LV1** at which the upper surface of the base layer **510** is positioned may be surrounded by the lower interposer protective layer **552a** and the plurality of interposer pads **570**. For example, a portion higher than the first level **LVI** of each of the interposer through electrodes **502** may be surrounded by the lower interposer protective layer **552a** and an interposer pad **570**. Among the plurality of interposer through electrodes **520**, the upper surface and side surface of the uppermost portion of the portion above the upper surface of the base layer **510** may be surrounded by a plurality of interposer pads **570**, and a side surface of the remaining portion (e.g., the other portion) may be surrounded by the lower interposer protective layer **552a**. For example, a top surface and an upper most portion of a side surface of each of the interposer through electrodes **520** may contact an interposer pad **570**, and a lower side surface of each of the interposer through electrode **502** at a vertical level above the first vertical level **LVI** may contact the lower interposer protective layer **552a**.

(55) A portion buried in the interposer protective layer **550a** among the plurality of interposer pads **570** may have a fourth thickness **T4**, and a portion that is not buried in the interposer protective layer **550a** and protrudes from the upper surface of the interposer protective layer **550a** may have a fifth thickness **T5** that is greater than the fourth thickness **T4**. For example, a portion of an interposer pad **570** disposed below the fourth vertical level **LV4** may have the fourth thickness **T4**, and a portion of the interposer pad **570** disposed above the fourth vertical level **LV4** may have the fifth thickness **T5**.

(56) The rear wiring protective layer **560** may cover all side surfaces of portions of the plurality of

interposer pads **570** protruding from/above the upper surface of the interposer protective layer **550a** and cover a portion of the upper surfaces of the plurality of interposer pads **570** but may have a terminal opening **5600** that exposes the remaining portions (e.g., the other portions) of the upper surfaces of the plurality of interposer pads **570** without covering the remaining portions.

(57) The interposer protective layer **550a** may further include an alignment recess **550Ra** vertically extending from the upper surface of the upper interposer protective layer **554a** toward the base layer **510**. A portion of the upper interposer protective layer **554a** may be exposed on the bottom surface of the alignment recess **550Ra**. The alignment recess **550Ra** extends from the upper surface of the upper interposer protective layer **554a** toward the base layer **510**, but may not completely penetrate the upper interposer protective layer **554a**. In some embodiments, the alignment recess **550Ra** may penetrate only an upper portion of the upper interposer protective layer **554a** and may not penetrate a lower portion of the upper interposer protective layer **554a**.

(58) The alignment recess **550Ra** may be filled by a rear wiring protective layer **560**. The alignment recess **550Ra** may be disposed to be horizontally spaced apart from the pad recess **570Ra**. The alignment recess **550Ra** may constitute/form an alignment key **AKa**. The alignment key **AKa** may be configured by an interface between the interposer protective layer **550a** and the rear wiring protective layer **560** extending into the interposer protective layer **550a**. For example, the alignment key **Aka** may be formed of the bottom surface and side surface of the alignment recess **550Ra**. For example, the alignment key **AKa** may extend vertically and/or horizontally along the bottom surface and side surfaces of the alignment recess **550Ra**. The bottom surface of the alignment recess **550Ra** and the bottom surface of the pad recess **570Ra** may be at the second vertical level **LV2**. For example, the bottom surface of the alignment recess **550** and the bottom surface of the pad recess **570Ra** may be at the same vertical level.

(59) FIG. **4** is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention. Specifically, **IVA** and **IVB** of FIG. **4** are enlarged cross-sectional views showing portions respectively corresponding to the regions **IIA** and **IIB** of FIG. **1**, and the same reference numerals as in FIG. **1** indicate the same components, and repeated descriptions thereof may be omitted.

(60) Referring to FIGS. **1** and **4**, an interposer **500b** includes a base layer **510**, a plurality of interposer through electrodes **520**, a wiring structure **530**, a plurality of micro pads **540**, an interposer protective layer **550b**, a plurality of interposer pads **570b**, and a rear wiring protective layer **560b**.

(61) The interposer protective layer **550b** may include a lower interposer protective layer **552b** covering the second surface **514** of the base layer **510** and an upper interposer protective layer **554b** covering the lower interposer protective layer **552b**. The interposer protective layer **550b** may be formed of an inorganic material. For example, the interposer protective layer **550b** may be formed of silicon oxide, silicon nitride, or a stacked structure of silicon oxide and silicon nitride. For example, the lower interposer protective layer **552b** may be made of oxide, and the upper interposer protective layer **554b** may be made of nitride.

(62) The interposer protective layer **550b** may have a pad recess **570Rb** vertically extending from an upper surface of the upper interposer protective layer **554b** toward the base layer **510**. A portion of the lower interposer protective layer **552b** may be exposed on the bottom surface of the pad recess **570Rb**.

(63) Spatially relative terms, such as “vertical,” “horizontal,” “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can

encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(64) The lower interposer protective layer **552b** and the upper interposer protective layer **554b** may have a first thickness **T1b** and a second thickness **T2b** as a maximum thickness, respectively. For example, portions of the lower interposer protective layer **552b** and the upper interposer protective layer **554b** that do not vertically overlap with the plurality of pad recesses **570Rb** and the plurality of interposer pads **570b** may respectively have the first thickness **T1b** and the second thickness **T2b** that are substantially constant. In some embodiments, the first thickness **T1b** may be less than the second thickness **T2b**. For example, the second thickness **T2b** may be about 0.5 μm to about 1.5 μm , and the first thickness **T1b** may be less than the second thickness **T2** and may be about 0.3 μm or more.

(65) A lower surface of the lower interposer protective layer **552b** may be at a first vertical level **LV1**, and a bottom surface of the pad recess **570Rb** may be at a second vertical level **LV2b** that is higher than the first vertical level **LV1**. The upper surface of the lower interposer protective layer **552b** and the lower surface of the upper interposer protective layer **554b** may contact each other at a third vertical level **LV3b** that is higher than the second vertical level **LV2b**. The upper surface of the upper interposer protective layer **554b** may be at a fourth vertical level **LV4** that is higher than the third vertical level **LV3b**.

(66) Each of the plurality of interposer pads **570b** may include a seed layer **572b** covering the bottom surface and side surfaces of the pad recess **570Rb** and a conductive pad layer **574b** on the seed layer **572b**. For example, the seed layer **572b** may cover the surface of the interposer protective layer **550b** exposed in the pad recess **570Rb**. The seed layer **572b** may conformally cover the bottom surface and side surfaces of the pad recess **570Rb**. At the side surface of the pad recess **570Rb**, an uppermost end of the seed layer **572b** may be at the fourth vertical level **LV4b**.

(67) Each of the plurality of interposer pads **570b** may fill the pad recess **570Rb** and protrude from the upper surface of the interposer protective layer **550b**. For example, the interposer pads **570b** may protrude from the upper surface of the upper interposer protective layer **554b**. For example, upper surfaces of the plurality of interposer pads **570b** may be at a vertical level that is higher than the fourth vertical level **LV4**. Lower surfaces of the plurality of interposer pads **570b** may be at the second vertical level **LV2b**.

(68) The plurality of interposer through electrodes **520** may extend into the plurality of interposer pads **570b**. The upper surface of the plurality of interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554b** may be at the fourth vertical level **LV4**. For example, the upper surfaces of the interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554b** may be at the same vertical level.

(69) The lowermost surfaces of the plurality of interposer pads **570b**, e.g., a portion excluding the portion in contact with the plurality of interposer through electrodes **520**, among the lower surfaces of the plurality of interposer pads **570b** may all be in contact with the lower interposer protective layer **552b**. For example, the lower surfaces of the plurality of interposer pads **570b** may contact and/or be covered by the plurality of interposer through electrodes **520** and the lower interposer protective layer **552b**.

(70) Among the plurality of interposer through electrodes **520**, a portion above the first vertical level **LV1** at which the upper surface of the base layer **510** is positioned may be surrounded by the lower interposer protective layer **552b** and the plurality of interposer pads **570b**. Among the plurality of interposer through electrodes **520**, the upper surface and side surface of the uppermost portion of the portion above the upper surface of the base layer **510** may be surrounded by a plurality of interposer pads **570b**, and a side surface of the remaining portion (e.g., the other portion) may be surrounded by the lower interposer protective layer **552b**.

(71) Each of the plurality of interposer pads **570b** may have a third thickness **T3b** as a maximum

thickness. The third thickness **T3b** may be about 2 μm to about 7 μm . A portion below the fourth vertical level **LV4** among the plurality of interposer pads **570b**, e.g., a portion buried in the interposer protective layer **550b**, may have a fourth thickness **T4**, and a portion above the fourth vertical level **LV4**, e.g., not buried in the interposer protective layer **550b**, and protruding from/above the upper surface of the interposer protective layer **550b** may have a fifth thickness **T5b** that is less than the fourth thickness **T4b**. For example, the fifth thickness **T5b** may be less than the fourth thickness **T4b**, but may have a value of about 4 μm or less.

(72) The rear wiring protective layer **560b** may have a terminal opening **560Ob** exposing a portion of the plurality of interposer pads **570b** and may cover the interposer protective layer **550b** and portions of the plurality of interposer pads **570b**. The rear wiring protective layer **560b** may cover all the sides of the portions of the plurality of interposer pads **570b** protruding from/above the upper surface of the interposer protective layer **550b** and cover a portion of the upper surface of the interposer pads **570b** but may have a terminal opening **5600b** that exposes the remaining portion (e.g., the other portion) of the upper surface of the interposer pads **570b**. The rear wiring protective layer **560b** may be made of an organic material. For example, the rear wiring protective layer **560b** may be formed of a polymer material.

(73) The rear wiring protective layer **560b** may have a sixth thickness **T6b** on the interposer protective layer **550b**. For example, the rear wiring protective layer **560b** may be substantially flat on most of the interposer protective layer **550b**. For example, on the interposer protective layer **550b**, the lowest thickness of the rear wiring protective layer **560b** may be the sixth thickness **T6b**, and the thickness of the rear wiring protective layer **560b** adjacent to the plurality of interposer pads **570b** on the interposer protective layer **550b** may be greater than the sixth thickness **T6b**. The sixth thickness **T6b** may be greater than the fifth thickness **T5b**. For example, the sixth thickness **T6b** may be about 3 μm to about 4.5 μm . The highest thickness of the rear wiring protective layer **560b** on the plurality of interposer pads **570b** may be a seventh thickness **T7b**, and the lowest thickness of the rear wiring protective layer **560b** on the interposer pads **570b** may be an eighth thickness **T8b**. For example, the seventh thickness **T7b** may be about 3 μm to about 4.5 μm , and the eighth thickness **T8b** may be about 1 μm to about 2.5 μm .

(74) The rear UBM layer **582** may conformally cover a portion of the upper surface of the interposer pad **570b** exposed by the terminal opening **5600b** and a portion of the rear wiring protective layer **560b** adjacent thereto.

(75) The interposer protective layer **550b** may further have an alignment recess **550Rb** vertically extending from the upper surface of the upper interposer protective layer **554b** toward the base layer **510**. A portion of the lower interposer protective layer **552b** may be exposed on the bottom surface of the alignment recess **550Rb**. The alignment recess **550Rb** may vertically extend from the upper surface of the upper interposer protective layer **554b** toward the base layer **510**, but may not completely penetrate the interposer protective layer **550b**. In some embodiments, the alignment recess **550Rb** may penetrate the upper interposer protective layer **554b**, but may penetrate only an upper portion of the lower interposer protective layer **552b** and may not penetrate a lower portion thereof. For example, the alignment recess **550 Rb** may expose the lower portion of the lower interposer protective layer **552b** at the bottom of the alignment recess **550 Rb**.

(76) The alignment recess **550Rb** may be filled by a rear wiring protective layer **560b**. The alignment recess **550Rb** may be disposed to be horizontally spaced apart from the pad recess **570Rb**. The alignment recess **550Rb** may constitute/form an alignment key **AKb**. The alignment key **AKb** may be configured by an interface between the interposer protective layer **550b** and the rear wiring protective layer **560b** extending into the interposer protective layer **550b**. For example, the alignment key **AKb** may be formed of the bottom surface and side surface of the alignment recess **550Rb**. For example, the alignment key **AKb** may extend vertically and/or horizontally along the bottom surface and side surfaces of the alignment recess **550Rb**. The bottom surface of the alignment recess **550Rb** and the bottom surface of the pad recess **570Rb** may be at the second

vertical level LV2b. For example, the bottom surface of the alignment recess 550Rb and the bottom surface of the pad recess 570Rb may be at the same vertical level.

(77) FIG. 5 is an enlarged cross-sectional view illustrating portions of an interposer according to an embodiment of the present invention. Specifically, VA and VB of FIG. 5 are enlarged cross-sectional views showing portions respectively corresponding to the regions IIA and IIB of FIG. 1, and the same reference numerals as in FIG. 1 indicate the same components, and repeated descriptions thereof may be omitted.

(78) Referring to FIGS. 1 and 5, an interposer 500c includes a base layer 510, a plurality of interposer through electrodes 520, a wiring structure 530, a plurality of micro pads 540, an interposer protective layer 550c, a plurality of interposer pads 570b, and a rear wiring protective layer 560b.

(79) The interposer protective layer 550c may include a lower interposer protective layer 552c covering the second surface 514 of the base layer 510 and an upper interposer protective layer 554c covering the lower interposer protective layer 552c. The interposer protective layer 550c may be formed of an inorganic material. For example, the lower interposer protective layer 552c may be made of oxide, and the upper interposer protective layer 554c may be made of nitride. The lower interposer protective layer 552c may protrude upward from the lower surface of the upper interposer protective layer 554c while surrounding the side surfaces of the plurality of interposer through electrodes 520, and may be in contact with lower surfaces of the plurality of interposer pads 570b. For example, a portion of the lower interposer protective layer 552c may be disposed at a higher vertical level than the lower surface of the upper interposer protective layer 554c.

(80) The interposer protective layer 550c may have a pad recess 570Rc extending from the upper surface of the upper interposer protective layer 554c toward the base layer 510. A portion of the upper interposer protective layer 554c and a portion of the lower interposer protective layer 552c may be exposed on the bottom surface of the pad recess 570Rc. A portion of the lower interposer protective layer 552c exposed on the bottom surface of the pad recess 570Rc may surround the plurality of interposer through electrodes 520.

(81) The lower interposer protective layer 552c may have a first thickness T1c below the upper interposer protective layer 554c, and the upper interposer protective layer 554c may have a second thickness T2c as a maximum thickness. A portion of the lower interposer protective layer 552c that contacts the lower surfaces of the plurality of interposer pads 570b and surrounds side surfaces of the plurality of interposer through electrodes 520 may have a thickness greater than the first thickness T1c in the vertical direction, and the remaining portion (e.g., the other portion) of the lower interposer protective layer 552c may have the first thickness T1c in a vertical direction. In some embodiments, a portion of the lower interposer protective layer 552c that contacts the lower surfaces of the plurality of interposer pads 570b and surrounds side surfaces of the plurality of interposer through electrodes 520 may have the first thickness T1c in a horizontal direction on side surfaces of the plurality of interposer through electrodes 520. For example, the horizontal thickness of the lower interposer protective layer 552c in the portion surrounding the interposer through electrodes 520 may be the same as the vertical thickness of the lower interposer protective layer 552c disposed below and/or vertically overlapping the upper interposer protective layer 554c.

(82) For example, the lower interposer protective layer 552c may have the first thickness T1c that is substantially constant in the vertical direction at a portion that overlaps with the upper interposer protective layer 554c in the vertical direction, and the upper interposer protective layer 554c may have a substantially constant second thickness T2c at portions that do not vertically overlap the plurality of pad recesses 570Rc and the plurality of interposer pads 570b. For example, the maximum thickness of the lower interposer protective layer 552c may be the same as the sum of the thickness of the upper interposer protective layer 554c and the thickness of the lower interposer protective layer 552c (e.g., the first thickness T1c) at the portion in which the plurality of interposer pads 570b overlap the lower and upper interposer protective layers 552c and 554c in the vertical

direction. In some embodiments, the first thickness **T1c** may be less than the second thickness **T2c**. For example, the second thickness **T2c** may be about 0.5 μm to about 1.5 μm , and the first thickness **T1c** may be less than the second thickness **T2c**, but may be about 0.3 μm or more.

(83) A lower surface of the lower interposer protective layer **552c** may be at a first vertical level **LV1**, and a bottom surface of the pad recess **570Rc** may be at a second vertical level **LV2b** that is higher than the first vertical level **LV1**. A portion of the upper surface of the lower interposer protective layer **552c** in contact with the lower surface of the upper interposer protective layer **554c** may be at a third vertical level **LV3c** that is lower than the second vertical level **LV2b**. The upper surface of the upper interposer protective layer **554c** may be at a fourth vertical level **LV4** that is higher than the second vertical level **LV2b**.

(84) Each of the plurality of interposer pads **570b** may include a seed layer **572b** covering the bottom surface and side surfaces of the pad recess **570Rc**, e.g., the surface of the interposer protective layer **550c** exposed in the pad recess **570Rc**, and a conductive pad layer **574b** on the seed layer **572b**.

(85) Each of the plurality of interposer pads **570b** may fill a pad recess **570Rc** and protrude from the upper surface of the interposer protective layer **550c**. For example, the interposer pads **570b** may protrude from the upper surface of the upper interposer protective layer **554c**. For example, upper surfaces of the plurality of interposer pads **570b** may be at a vertical level that is higher than the fourth vertical level **LV4**. Lower surfaces of the plurality of interposer pads **570b** may be at the second vertical level **LV2b**.

(86) A portion buried in and/or below the interposer protective layer **550c** among the plurality of interposer pads **570b** may have a fourth thickness **T4b**, and a portion that is not buried in and/or above the interposer protective layer **550c** and protrudes from the upper surface of the interposer protective layer **550c** may have a fifth thickness **T5b** that is less than the fourth thickness **T5b**.

(87) The plurality of interposer through electrodes **520** may extend into the plurality of interposer pads **570b**. The upper surfaces of the plurality of interposer through electrodes **520** and the upper surface of the upper interposer protective layer **554c** may be at the fourth vertical level **LV4**. For example, the upper surfaces of the interposer through electrodes **520** and the upper surface of the interposer protective layer **550c** may be at the same vertical level. In certain embodiments, top surfaces of the interposer through electrodes **520** and a top surface of the interposer protective layer **550c** may be at different vertical levels as shown in FIGS. 4 and 5.

(88) The lowermost surfaces of the plurality of interposer pads **570b**, e.g., a portion of the lower surface of the plurality of interposer pads **570b** excluding a portion in contact with the plurality of interposer through electrodes **520**, may be in contact with the upper interposer protective layer **554c** and the lower interposer protective layer **552c**. For example, among the lowermost surfaces of the plurality of interposer pads **570**, a portion adjacent to the plurality of interposer through electrodes **520** may be in contact with the lower interposer protective layer **552c**, and the remaining portions (e.g., the other portions) may be in contact with the upper interposer protective layer **554c**. For example, the lower surfaces of the plurality of interposer pads **570b** may be all covered by and/or contact the plurality of interposer through electrodes **520**, the upper interposer protective layer **554c**, and the lower interposer protective layer **552c**.

(89) Among the plurality of interposer through electrodes **520**, a portion above the first vertical level **LVI** at which the upper surface of the base layer **510** is positioned may be surrounded by the lower interposer protective layer **552c** and the plurality of interposer pads **570b**. Among the plurality of interposer through electrodes **520**, the upper surface and side surface of the uppermost portion of the portion above the upper surface of the base layer **510** may be surrounded by a plurality of interposer pads **570b**, and a side surface of the remaining portion above the upper surface of the base layer **510** may be surrounded by the lower interposer protective layer **552c**.

(90) The rear wiring protective layer **560b** may have a terminal opening **560Ob** exposing a portion of the plurality of interposer pads **570b** and may cover the interposer protective layer **550c** and

portions of the plurality of interposer pads **570b**.

(91) The interposer protective layer **550c** may further have an alignment recess **550Rc** vertically extending from the upper surface of the upper interposer protective layer **554c** toward the base layer **510**. A portion of the upper interposer protective layer **554c** may be exposed on the bottom surface of the alignment recess **550Rc**. The alignment recess **550Rc** may vertically extend from the upper surface of the upper interposer protective layer **554c** toward the base layer **510**, but may not completely penetrate the upper interposer protective layer **550c**. In some embodiments, the alignment recess **550Rc** may penetrate only an upper portion of the upper interposer protective layer **554c** and may not penetrate a lower portion of the upper interposer protective layer **554c**.

(92) The alignment recess **550Rc** may be filled by a rear wiring protective layer **560b**. The alignment recess **550Rc** may be disposed to be horizontally spaced apart from the pad recess **570Rc**. The alignment recess **550Rc** may constitute/form an alignment key **AKc**. The alignment key **AKc** may be configured by an interface between the interposer protective layer **550c** and the rear wiring protective layer **560b** extending into the interposer protective layer **550c**. For example, the alignment key **AKc** may be formed of the bottom surface and side surface of the alignment recess **550Rc**. For example, the alignment key **AKc** may vertically and horizontally extend along the bottom surface and side surfaces of the alignment recess **550Rc**. The bottom surface of the alignment recess **550Rc** and the bottom surface of the pad recess **570Rb** may be at the second vertical level **LV2b**. For example, the bottom surface of the alignment recess **550Rc** and the bottom surface of the pad recess **570Rb** may be at the same vertical level.

(93) FIGS. **6A** to **6G** are cross-sectional views illustrating a method of manufacturing an interposer according to an embodiment of the present invention in stages. Specifically, FIGS. **6A** to **6G** are cross-sectional views illustrating a method of manufacturing the interposer shown in FIGS. **1** and **2** in stages.

(94) Referring to FIG. **6A**, after preparing the base layer **510**, a plurality of interposer through electrodes **520** penetrating the base layer **510** from the first surface **512** to the second surface **514** of the base layer **510**, a wiring structure **530** disposed on the first surface **512** of the base layer **510**, and a plurality of micro pads **540** on the first surface **512** of the base layer **510** and electrically connected to the wiring structure **530** are formed.

(95) The base layer **510** on which the plurality of interposer through electrodes **520**, the wiring structure **530**, and the plurality of micro pads **540** are formed may be attached to a support substrate such that the first surface **512** faces the support substrate. A release adhesive film may be interposed between the base layer **510** and the support substrate and the base layer **510** may be attached to the support substrate by the release adhesive film. After attaching the base layer **510** to the support substrate, by removing a portion opposite to the first surface **512** of the base layer **510**, a plurality of interposer through electrodes **520** may be formed to protrude outward from the second surface **514** of the base layer **510**. For example, an upper portion of the base layer **510** may be removed to form the second surface **514** of the base layer **510** so that the second surface **514** may be at a lower vertical level than top surfaces of the interposer through electrodes **520**.

(96) On the second surface **514** of the base layer **510**, by sequentially forming a preliminary lower interposer **552P** that conformally covers the second surface **514** of the base layer **510** and the side surfaces and upper surfaces of the plurality of interposer through electrodes **520** protruding outward from the second surface **514** of the base layer **510**, a preliminary upper interposer protective layer **554P**, and a cover interposer protective layer **556P**, a preliminary interposer protective layer **550P** may be formed. For example, the preliminary interposer protective layer **550P** may include the preliminary lower interposer protective layer **552P**, the preliminary upper interposer protective layer **554P**, and the cover interposer protective layer **556P**. The preliminary lower interposer protective layer **552P** may be formed to have a first thickness **T1**, and the preliminary upper interposer protective layer **554P** may be formed to have a thickness greater than the second thickness **T2** of FIG. **6B**.

(97) Each of the preliminary lower interposer protective layer **552P**, the preliminary upper interposer protective layer **554P**, and the cover interposer protective layer **556P** may be formed of an inorganic material. For example, the preliminary lower interposer protective layer **552P**, the preliminary upper interposer protective layer **554P**, and the cover interposer protective layer **556P** may be respectively formed of silicon oxide, silicon nitride, and silicon oxide.

(98) Referring to FIGS. **6A** and **6B** together, until all of the cover interposer protective layers **556P** are removed and the thickness of the preliminary upper interposer protective layer **554P** becomes the second thickness **T2**, a planarization process is performed in which upper portions of the preliminary interposer protective layer **550P** and the plurality of interposer through electrodes **520** are removed.

(99) Referring to FIGS. **6B** and **6C** together, after forming a first mask pattern **MKI** covering a portion of the preliminary interposer protective layer **550P** and not covering the plurality of interposer through electrodes **520**, by removing a portion of the preliminary interposer protective layer **550P** using the first mask pattern **MKI** as an etching mask, an interposer protective layer **550** having a pad recess **570R** and an alignment recess **550R** spaced apart from each other may be formed. For example, the interposer protective layer **550** may include a lower interposer protective layer **552** and an upper interposer protective layer **554**. The interposer protective layer **550** may be formed such that a portion of the lower interposer protective layer **552** is exposed on the bottom surface of each of the pad recess **570R** and the alignment recess **550R**.

(100) A lower surface of the lower interposer protective layer **552** may be at a first vertical level **LVI**, and a bottom surface of the pad recess **570R** may be formed to be positioned at a second vertical level **LV2** that is higher than the first vertical level **LV1**. The upper surface of the lower interposer protective layer **552** and the lower surface of the upper interposer protective layer **554** may contact each other at a third vertical level **LV3** that is higher than the second vertical level **LV2**. An upper surface of each of the upper interposer protective layer **554** and the plurality of interposer through electrodes **520** may be formed to be positioned at a fourth vertical level **LV4** that is higher than the third vertical level **LV3**. For example, the upper surface of the upper interposer protective layer **554** and upper surfaces of the interposer through electrodes **520** may be at the same vertical level.

(101) After the pad recess **570R** and the alignment recess **550R** are formed in the interposer protective layer **550**, the first mask pattern **MKI** may be removed.

(102) Referring to FIG. **6D**, a preliminary seed layer **572P** conformally covering exposed surfaces of the interposer protective layer **550** and the plurality of interposer through electrodes **520** is formed. The preliminary seed layer **572P** may be formed to conformally cover the uppermost surface of the interposer protective layer **550**, the bottom surface and side surfaces of each of the pad recess **570R** and the alignment recess **550R**, and the upper surfaces and the side surfaces of the plurality of interposer through electrodes **520** exposed in the pad recess **570R**.

(103) Referring to FIG. **6E**, a second mask pattern **MK2** covering the preliminary seed layer **572P** is formed. The second mask pattern **MK2** may be formed to cover portions of the preliminary seed layer **572P** including a portion of the preliminary seed layer **572P** that vertically overlaps with the alignment recess **550R** without covering a portion of the preliminary seed layer **572P** that vertically overlaps with the pad recess **570R**. For example, the second mask pattern **MK2** may expose a portion of the preliminary seed layer **572P** vertically overlapping the pad recess **570R**.

(104) Thereafter, a conductive pad layer **574** is formed on the preliminary seed layer **572P** that is not covered by the second mask pattern **MK2** and is exposed. The conductive pad layer **574** may be formed using a plating method using the preliminary seed layer **572P** as a seed. In some embodiments, the conductive pad layer **574** may be formed using an electrolytic plating method using the preliminary seed layer **572P** as a seed. The conductive pad layer **574** may be formed so that its upper surface is positioned at a higher vertical level than the fourth vertical level **LV4**.

(105) Referring to FIGS. **6E** and **6F** together, after removing the second mask pattern **MK2**, a

portion of the preliminary seed layer **572P** that is not covered and exposed by the conductive pad layer **574** is removed, so that an interposer pad **570** including a seed layer **572** and a conductive pad layer **574** is formed. The interposer pad **570** may be formed to have a third thickness **T3** as a maximum thickness.

(106) A portion of the interposer pad **570** disposed below the fourth vertical level **LV4**, e.g., a portion buried in the interposer protective layer **550**, may be formed to have a fourth thickness **T4**, and a portion of the interposer pad **570** disposed above the fourth vertical level **LV4**, e.g., the portion that is not buried in the interposer protective layer **550** and protrudes from/above the upper surface of the interposer protective layer **550**, may be formed to have a fifth thickness **T5** greater than the fourth thickness **T4**.

(107) Referring to FIG. **6G**, a rear wiring protective layer **560** that has a terminal opening **5600** exposing a portion of the plurality of interposer pads **570** and covers the remaining portions (e.g., the other portions) of the plurality of interposer pads **570** and the interposer protective layer **550** is formed. The rear wiring protective layer **560** may be formed to cover all side surfaces of portions of the plurality of interposer pads **570** protruding from the upper surface of the interposer protective layer **550** and cover a portion of the upper surface of the interposer pads **570**, but to have a terminal opening **5600** not covering and exposing the remaining portion (e.g., the other portion) of the upper surface. For example, the terminal opening may expose a top surface of an interposer pad **570**.

(108) The rear wiring protective layer **560** may be formed to have a sixth thickness **T6** on the interposer protective layer **550**. For example, the rear wiring protective layer **560** may have a sixth thickness **T6** greater than the fifth thickness **T5** as the lowest thickness among portions formed on and/or vertically overlapping the interposer protective layer **550**, and a portion adjacent to the plurality of interposer pads **570** formed on and/or vertically overlapping the interposer protective layer **550** may be formed to have a value greater than the sixth thickness **T6**. The rear wiring protective layer **560** may be formed to have a seventh thickness **T7** as the highest thickness and an eighth thickness **T8** as the lowest thickness among portions formed on and/or vertically overlapping the plurality of interposer pads **570**.

(109) The alignment recess **550R** may be filled by the rear wiring protective layer **560** to constitute/form the alignment key **AK** configured by an interface of the interposer protective layer **550** and the rear wiring protective layer **560** extending into the interposer protective layer **550**, e.g., along the bottom surface and side surfaces of the alignment recess **550R**.

(110) Then, as shown in FIGS. **1** and **2**, by forming an interposer connection terminal **580** including and/or formed of a rear UBM layer **582** on interposer pad **570** and an interposer conductive cap **584** on the rear UBM layer **582**, an interposer **500** may be formed.

(111) Referring to FIGS. **1**, **2**, and **6A** to **6G**, in relation to the interposer **500** according to an embodiment of the present invention, since a lower portion of the interposer pad **570** is formed to fill the pad recess **570R**, the thickness of the interposer pad **570** may be increased, e.g., without increasing thicknesses of interposers **500**. Thick interposer pads **570** may be beneficial for preventing warpage in the interposer **500**. Since the manufacturing method of the interposer **500** according to certain embodiments of the present invention does not perform a separate process for forming the alignment key **AK**, manufacturing cost of the interposer **500** may be reduced by reducing the process steps.

(112) FIGS. **7A** to **7D** are cross-sectional views illustrating a method of manufacturing an interposer according to an embodiment of the present invention in stages. Specifically, FIGS. **7A** to **7D** are cross-sectional views illustrating a method of manufacturing the interposer shown in FIG. **3** in stages. In the description of FIGS. **7A** to **7D**, the content overlapping with the description of FIGS. **6A** to **6G** may be omitted.

(113) Referring to FIG. **7A**, on the second surface **514** of the base layer **510**, by sequentially forming a preliminary lower interposer **552Pa** that conformally covers the second surface **514** of

the base layer **510** and the side surfaces and upper surfaces of the plurality of interposer through electrodes **520** protruding outward from the second surface **514** of the base layer **510**, a preliminary upper interposer protective layer **554Pa**, and a cover interposer protective layer **556Pa**, a preliminary interposer protective layer **550Pa** may be formed. For example, the preliminary interposer protective layer **550Pa** may include the preliminary lower interposer protective layer **552Pa**, the preliminary upper interposer protective layer **554Pa**, and the cover interposer protective layer **556Pa**. The preliminary lower interposer protective layer **552Pa** may be formed to have a first thickness **T1a**, and the preliminary upper interposer protective layer **554Pa** may be formed to have a thickness greater than the second thickness **T2a** of FIG. **7B**.

(114) Referring to FIGS. **7A** and **7B** together, until all of the cover interposer protective layers **556Pa** are removed and the thickness of the preliminary upper interposer protective layer **554Pa** becomes the second thickness **T2a**, a planarization process is performed in which upper portions of the preliminary interposer protective layer **550Pa** and the plurality of interposer through electrodes **520a** are removed.

(115) Referring to FIGS. **7B** and **7C** together, by removing a portion of the preliminary interposer protective layer **550Pa**, an interposer protective layer **550a** may be formed. For example, the interposer protective layer **550a** may include a pad recess **570Ra** and an alignment recess **550Ra** horizontally spaced apart from each other. The interposer protective layer **550a** may include a lower interposer protective layer **552a** and an upper interposer protective layer **554a**. The interposer protective layer **550a** may be formed such that a portion of the upper interposer protective layer **554a** is exposed on the bottom surface of each of the pad recess **570Ra** and the alignment recess **550Ra**.

(116) A lower surface of the lower interposer protective layer **552a** may be at a first vertical level **LV1**, and a bottom surface of the pad recess **570Ra** may be formed to be positioned at a second vertical level **LV2** that is higher than the first vertical level **LV1**. The upper surface of the lower interposer protective layer **552a** and the lower surface of the upper interposer protective layer **554a** may contact each other at a third vertical level **LV3a** that is lower than the second vertical level **LV2**. An upper surface of each of the upper interposer protective layer **554a** and the plurality of interposer through electrodes **520** may be formed to be positioned at a fourth vertical level **LV4** that is higher than the second vertical level **LV2**. For example, top surfaces of the interposer through electrodes **520** and a top surface of the upper interposer protective layer **554a** may be formed to be at the same vertical level.

(117) Referring to FIG. **7D**, with reference to the description of embodiments illustrated in FIGS. **6D** to **6F**, an interposer pad **570** including a seed layer **572** and a conductive pad layer **574** is formed.

(118) The interposer pad **570** may be formed to have a third thickness **T3** as a maximum thickness, and a portion of the interposer pad **570** below the fourth vertical level **LV4**, e.g., a portion buried in the interposer protective layer **550a**, may have a fourth thickness **T4**, and a portion above the fourth vertical level **LV4**, e.g., a portion protruding from/above the upper surface of the interposer protective layer **550a** without being embedded in the interposer protective layer **550a** may have a fifth thickness **T5** greater than the fourth thickness **T4**. In some embodiments, the fifth thickness **T5** may be less than the fourth thickness **T4**.

(119) Thereafter, as shown in FIG. **3**, the interposer **500a** may be formed by forming the rear wiring protective layer **560** and the interposer connection terminal **580**.

(120) The interposer **500b** shown in FIG. **4** and the interposer **500c** shown in FIG. **5B** may be formed by respectively forming the pad recess **570Rb** shown in FIG. **4** and the pad recess **570Rc** shown in FIG. **4** deeper than the depth of the pad recess **570R** of FIG. **2** and the pad recess **570Ra** of FIG. **3** described through FIGS. **6A** to **7D**, so that a detailed description thereof will be omitted.

(121) FIG. **8** is a cross-sectional view illustrating a semiconductor package having an interposer according to an embodiment of the present invention.

(122) Referring to FIG. 8, a semiconductor package **1000** may include a main board **600** on which an interposer **500** is mounted, at least one sub-semiconductor package **10** including a first semiconductor chip **100** and a plurality of second semiconductor chips **200** and attached to the interposer **500**, and a third semiconductor chip **400**. The at least one sub-semiconductor package **10** and the third semiconductor chip **400** may be mounted on the interposer **500** while being spaced apart from each other in a horizontal direction.

(123) The at least one sub-semiconductor package **10** and the third semiconductor chip **400** may be electrically connected to the interposer **500** by a plurality of micro pads **540**. The at least one sub-semiconductor package **10** may have a plurality of first upper surface connection pads **122**, and the third semiconductor chip **400** may have a plurality of second upper surface connection pads **420**. The plurality of micro pads **540** may be electrically connected to the plurality of first upper surface connection pads **122** and the plurality of second upper surface connection pads **420**. In some embodiments, chip connection bumps may be disposed on the plurality of micro pads **540**, and the micro pads **540** may be electrically connected to the plurality of first upper surface connection pads **122** and the plurality of second upper surface connection pads **420** through the chip connection bumps interposed therebetween.

(124) The sub-semiconductor package **10** may include a first semiconductor chip **100** and a plurality of second semiconductor chips **200**. Although the sub-semiconductor package **10** is illustrated as including four second semiconductor chips **200** in FIG. 8, the present invention is not limited thereto. For example, the sub-semiconductor package **10** may include two or more second semiconductor chips **200**. In some embodiments, the sub-semiconductor package **10** may include a multiple of four second semiconductor chips **200**. The plurality of second semiconductor chips **200** may be sequentially stacked on the first semiconductor chip **100** in a vertical direction. Each of the first semiconductor chip **100** and the plurality of second semiconductor chips **200** may have an active surface facing downward.

(125) The first semiconductor chip **100** may include a first semiconductor substrate **110** having a first semiconductor device formed on an active surface thereof, a first upper surface connection pad **122** and a first lower surface connection pad **124** respectively disposed on an active surface and an inactive surface of the first semiconductor substrate **110**, and a first through electrode **130** penetrating at least a portion of the first semiconductor substrate **110** to electrically connect the first upper surface connection pad **122** and the first lower surface connection pad **124**. For example, the inactive surface of the first semiconductor substrate **110** may be a surface opposite the active surface of the first semiconductor substrate **110**.

(126) The first semiconductor substrate **110** may include, for example, a semiconductor material such as silicon (Si). Alternatively, the first semiconductor substrate **110** may include a semiconductor element such as germanium (Ge), or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP). The first semiconductor substrate **110** may include a conductive region, for example, a well doped with impurities. The first semiconductor substrate **110** may have various device isolation structures such as a shallow trench isolation (STI) structure.

(127) In this specification, an upper surface and a lower surface of a semiconductor substrate such as the first semiconductor substrate **110** refer to an active surface side and an inactive surface side of the semiconductor substrate, respectively. For example, even when the active surface of the semiconductor substrate is positioned below the inactive surface in the final product, in this specification, the active surface side of the semiconductor substrate is referred to as an upper surface of the semiconductor substrate, and the inactive surface side of the semiconductor substrate is referred to as a lower surface. Also, the terms upper surface and lower surface may be respectively used for each component disposed on the active surface and the component disposed on the inactive surface of the semiconductor substrate.

(128) The first semiconductor device including a plurality of individual devices of various types

may be formed on the active surface of the first semiconductor substrate **110**.

(129) Each of the second semiconductor chips **200** may include a second semiconductor substrate **210** having a second semiconductor device formed on the active surface, an inner upper surface connection pad **222** and an inner lower surface connection pad **224** respectively disposed on the active and inactive surfaces of the second semiconductor substrate **210**, and a second through electrode **230** penetrating at least a portion of the second semiconductor substrate **210** to electrically connect the inner upper surface connection pad **222** and the inner lower surface connection pad **224**.

(130) The second semiconductor substrate **210**, the inner upper surface connection pad **222**, the inner lower surface connection pad **224**, and the second through electrode **230** may be substantially the same as the first semiconductor substrate **110**, the first upper surface connection pad **122**, the first bottom connection pad **124**, and the first through electrode **130**, respectively, and detailed descriptions thereof will be omitted.

(131) An inner connection terminal **240** may be attached to the inner upper surface connection pad **222** of each of the plurality of second semiconductor chips **200**. Inner connection terminals **240** may electrically connect first lower surface connection pads **124** of the first semiconductor chip **100** and inner upper surface connection pads **222** of the plurality of second semiconductor chips **200**, and inner lower surface connection pads **224** and inner upper surface connection pads **222** of each of the plurality of second semiconductor chips **200**.

(132) In some embodiments, the first semiconductor chip **100** may include a serial-parallel conversion circuit and may be a buffer chip for controlling a DRAM semiconductor chip. In some embodiments, each of the second semiconductor chip **200** may be a DRAM semiconductor chip. The first semiconductor chip **100** may be a master chip, and each of the second semiconductor chips **200** may be a slave chip, and the sub-semiconductor package **10** including the first semiconductor chip **100** and the plurality of second semiconductor chips **200** may be an HBM DRAM.

(133) An insulating adhesive layer **350** may be disposed between each pair of the first semiconductor chip **100** and the plurality of second semiconductor chips **200**. The insulating adhesive layer **350** may include a non-conductive film (NCF), a non-conductive paste (NCP), an insulating polymer, or an epoxy resin. The insulating adhesive layer **350** may surround the inner connection terminals **240** and may fill spaces between the first semiconductor chip **100** and the plurality of second semiconductor chips **200**, respectively.

(134) In some embodiments, among the plurality of second semiconductor chips **200**, the second semiconductor chip **200** disposed at the uppermost position farthest from the first semiconductor chip **100** may not include the inner lower surface connection pad **224** and the second through electrode **230**. In some embodiments, among the plurality of second semiconductor chips **200**, the thickness of the second semiconductor chip **200** positioned at the uppermost position farthest from the first semiconductor chip **100** may be greater than the thicknesses of the remaining second semiconductor chips **200**.

(135) The width and area of the first semiconductor chip **100** may be greater than the width and area of each of the plurality of second semiconductor chips **200**. The sub-semiconductor package **10** may further include a molding layer **300** surrounding the side surfaces of the plurality of second semiconductor chips **200** and the side surfaces of the insulating adhesive layer **350** on the first semiconductor chip **100**. The molding layer **300** may include and/or be formed of, for example, an epoxy mold compound (EMC).

(136) The third semiconductor chip **400** may include a third semiconductor substrate **410** and a second upper surface connection pad **420**. The third semiconductor substrate **410** and the second upper surface connection pad **420** may be components substantially the same as or similar to the first semiconductor substrate **110** and the first upper surface connection pad **122**, respectively, or may be components substantially the same or similar to the second semiconductor substrate **120**

and the inner upper surface connection pad **222**, respectively, so that a detailed description thereof will be omitted.

(137) The third semiconductor chip **400** may be, for example, a central processing unit (CPU) chip, a graphics processing unit (GPU) chip, or an application processor (AP) chip.

(138) The interposer **500** may include a base layer **510**, a plurality of interposer through electrodes **520**, a wiring structure **530**, a plurality of micro pads **540**, an interposer protective layer **550**, a plurality of interposer pads **570**, and a rear wiring protective layer **560**.

(139) In some embodiments, the semiconductor package **1000** may include any one of the interposers **500a**, **500b**, and **500c** described with reference to FIGS. **3** to **5** instead of the interposer **500**. The interposers **500**, **500a**, **500b**, and **500c** have been described in detail with reference to FIGS. **1** to **7D**, and overlapping descriptions will be omitted.

(140) A first underfill layer **380** may be disposed between the sub-semiconductor package **10** and the interposer **500**, and a second underfill layer **480** may be disposed between the third semiconductor chip **400** and the interposer **500**.

(141) The semiconductor package **1000** may further include a package molding layer **900** surrounding the sub semiconductor package **10** and the side surfaces of the third semiconductor chip **400** on the interposer **500**. The package molding layer **900** may include and/or be formed of, for example, an EMC.

(142) In some embodiments, the package molding layer **900** may cover the upper surface of the interposer **500** and side surfaces of each of the sub-semiconductor package **10** and the third semiconductor chip **400**, but may not cover the upper surfaces of the sub-semiconductor package **10** and the third semiconductor chip **400**, and the semiconductor package **1000** may further include a heat dissipation member covering upper surfaces of the sub-semiconductor package **10** and the third semiconductor chip **400**. The heat dissipation member may include a heat dissipation plate such as a heat slug or a heat sink. In some embodiments, the heat dissipation member may surround the sub-semiconductor package **10**, the third semiconductor chip **400**, and the interposer **500** on the upper surface of the package base substrate **600**. For example, the heat dissipation member may vertically overlap the whole bodies of sub-semiconductor package **10**, the third semiconductor chip **400**, and the interposer **500**.

(143) A plurality of interposer connection terminals **580** may be attached to the plurality of interposer pads **570**. The plurality of interposer connection terminals **580** may electrically connect the interposer **500** and the package base substrate **600**.

(144) The package base substrate **600** may include a base board layer **610**, a board upper surface pad **622** and a board lower surface pad **624** respectively disposed on the upper surface and the lower surface of the base board layer **610**, and a board wiring layer **630** that electrically connects the board upper surface pad **622** and the board lower surface pad **624** in the base board layer **610**. In some embodiments, the package base board **600** may be a printed circuit board. For example, the package base board **600** may be a multi-layer printed circuit board. The base board layer **610** may be made of at least one material selected from phenol resin, epoxy resin, and polyimide.

(145) A solder resist layer (not shown) exposing the board upper surface pad **622** and the board lower surface pad **624** may be formed on each of the upper and lower surfaces of the base board layer **610**. The interposer connection terminal **580** may be electrically connected to and/or contact the board lower surface pad **624**, and the package connection terminal **650** may be electrically connected to and/or contact the board upper pad **622** on the bottom of the board. The interposer connection terminal **540** may electrically connect the plurality of pad wiring layers **524** and the board upper surface pad **622**. The package connection terminal **650** electrically connected to and/or contact the board upper surface pad **622** may electrically connect the semiconductor package **1000** to the outside.

(146) Since the semiconductor package **1000** according to an embodiment of the present invention includes the interposer **500** that may prevent a warpage phenomenon from occurring, electrical

connection reliability between the sub-semiconductor package **10**, the third semiconductor chip **400** and the interposer **500** may be improved, so that operation reliability of the semiconductor package **1000** may be improved.

(147) While the inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. An interposer manufacturing method comprising: preparing a base layer having a first surface and a second surface opposite to each other, and an interposer through electrode penetrating the base layer and protruding from the second surface of the base layer; forming an interposer protective layer covering the second surface of the base layer and having an upper surface at the same vertical level as an upper surface of the interposer through electrode; removing a portion of the interposer protective layer to form a pad recess in which an upper portion of the interposer through electrode is exposed and a portion of the interposer protective layer is exposed on a bottom surface of the pad recess; forming an interposer pad filling the pad recess of the interposer protective layer and protruding from the interposer protective layer; and forming a rear wiring protective layer having a terminal opening exposing a portion of the interposer pad and covering the remaining portion of the interposer pad and the interposer protective layer.
2. The interposer manufacturing method of claim 1, further comprising forming an alignment recess horizontally spaced apart from the pad recess and exposing a portion of a protective layer of the interposer on the bottom surface.
3. The interposer manufacturing method of claim 2, wherein a bottom surface of the pad recess and a bottom surface of the alignment recess are formed at the same vertical level.
4. The interposer manufacturing method of claim 2, wherein the rear wiring protective layer fills the alignment recess.
5. The interposer manufacturing method of claim 1, wherein a thickness of the remaining portion of the interposer pad protruding from the interposer protective layer is greater than a thickness of a portion of the interposer pad filling the pad recess of the interposer protective layer.
6. The interposer manufacturing method of claim 1, wherein the forming of the interposer protective layer comprises: forming a lower interposer protective layer covering the second surface of the base layer; and forming an upper interposer protective layer on the lower interposer protective layer, the upper interposer protective layer including a material different from that of the lower interposer protective layer.
7. The interposer manufacturing method of claim 6, wherein the forming of the pad recess comprises exposing a portion of the lower interposer protective layer on a bottom surface of the pad recess, wherein the whole lower surface of the interposer pad contacts the interposer through electrode and the lower interposer protective layer.
8. The interposer manufacturing method of claim 6, wherein the forming of the pad recess comprises: exposing a portion of the upper interposer protective layer and a portion of the lower interposer protective layer on the bottom surface of the pad recess; and surrounding a side surface of the interposer through electrode with a portion of the lower interposer protective layer exposed on the bottom surface of the pad recess.
9. The interposer manufacturing method of claim 6, wherein a maximum thickness of the lower interposer protective layer is greater than a maximum thickness of the upper interposer protective layer.
10. The interposer manufacturing method of claim 1, wherein the interposer pad covers all of the upper portion of the interposer through electrode exposed in the pad recess.
11. The interposer manufacturing method of claim 1, wherein the interposer protective layer is

formed directly on the second surface of the base layer.

12. The interposer manufacturing method of claim 6, wherein the forming of the pad recess comprises exposing removing a portion of the upper interposer protective layer to expose a portion of the lower interposer protective layer and removing a portion of the exposed interposer protective layer, wherein the interposer through electrodes pass through the lower interposer protective layer and a portion of the upper interposer protective layer is disposed over the interposer through electrodes.

13. The interposer manufacturing method of claim 1, wherein the thickness of a portion of the interposer pad above the interposer through electrodes is less than a thickness of a portion of the interposer pad that is not over the interposer through electrodes.
